

UPPER-BODY CONTEXT AS REQUIRED INTERPRETATION LAYER FOR INTIMATE PHYSIOLOGY: STATE, ARTIFACT CONTROL, AND CONFIDENCE CONDITIONING (NON-LIMITING)

A. Technical Field

This section relates to systems that measure intimate-region physiology using one or more intimate-focused nodes (Nodes 1–4) and require an upper-body context node (Node 5) to provide physiologic state, motion state, thermal coupling state, and wear integrity necessary for correct interpretation of intimate-region signals.

B. Problem Addressed (Non-Limiting)

Intimate-region physiology signals can be highly state-dependent. Similar local readings may correspond to different underlying conditions depending on whole-body context, including activity, stress/activation, temperature shifts, and motion artifacts. Without upper-body context, intimate signals may be misinterpreted, improperly compared across time, or incorrectly correlated across devices. In various embodiments, Node 5 supplies a systemic truth layer that enables intimate nodes to be interpreted as patterned physiology rather than ambiguous local snapshots.

C. Summary (Non-Limiting)

In various embodiments, Node 5 is an upper-body wearable implemented as a chest-mounted device, ring, wristband, or equivalent, configured to capture systemic context signals including: cardiac electrical sensing (ECG electrodes) and derived HR/HRV features, optical pulse sensing (PPG) and pulse-derived features, inertial sensing (6-axis IMU) for motion state and artifact detection, respiratory dynamics (derived from motion, impedance, or equivalent), electrodermal activity (EDA) / skin conductance for autonomic activation context, temperature/thermal context for systemic state and coupling plausibility, and contact integrity / wear detection for trust scoring. Node 5 generates privacy-preserving context tags and confidence indicators that are used to (i) gate validity of intimate-node windows, (ii) normalize interpretation of intimate signals by systemic state, and (iii) reduce false inferences caused by motion, stress, temperature drift, or poor coupling.

D. Why Upper-Body Context is System-Critical for Intimate Data (Non-Limiting)

In various embodiments, Node 5 is important to the system because it provides one or more of the following indispensable functions: 1. State Disambiguation: Node 5 provides systemic state tags (e.g., rest, active, sleep, autonomic activation) used to distinguish whether changes seen in Nodes 1–4 reflect meaningful physiologic change versus state-driven variation. 2. Artifact Rejection and Trust Scoring: Node 5 detects motion state and wear integrity to determine when intimate-node data is likely corrupted by movement, posture shifts, partial coupling, or transient disturbances, enabling the system to suppress, exclude, or down-weight those windows. 3. Normalization and Comparability Over Time: Node 5 enables like-for-like comparisons by selecting windows with comparable systemic context (e.g., rest-state windows vs. active-state windows), improving longitudinal interpretation of intimate-region trends. 4. Thermal and Autonomic Context Conditioning: Node 5 provides thermal context and autonomic activation context (via HRV/EDA features) to condition how intimate signals are interpreted, since vascular and tissue-related signals can vary substantially with stress/activation and temperature. 5. System Confidence

Indicators: Node 5 produces per-window confidence indicators that the system uses to label outputs from Nodes 1–4 with explicit reliability measures, preventing overconfident conclusions from low-quality windows.

E. Node 5 Context Outputs Used by the System (Non-Limiting)

In various embodiments, Node 5 produces one or more context outputs including: 1. Motion State Tag (still / light movement / active / high-motion) 2. Wear Integrity Index (contact integrity + coupling plausibility) 3. Autonomic Activation Tag (confidence-scored, derived from HRV and/or EDA) 4. Thermal Context Tag (temperature trends, coupling stability, drift plausibility) 5. Respiration Tag (rate/variability confidence, when available) 6. Composite Context Vector used to gate and condition intimate-node interpretation

F. Conditioning of Intimate Nodes 1–4 by Upper-Body Context (Non-Limiting)

In various embodiments, Node 5 context outputs condition intimate nodes as follows: Node 1 (all-day/night penile physiology): Node 5 supports rest/sleep classification and motion gating so nocturnal or resting patterns can be interpreted with reduced artifact and improved comparability. Node 2 (intravaginal longitudinal physiology): Node 5 provides autonomic and thermal context that conditions impedance/hydration/pH-related interpretations and reduces false trend inference during systemic activation or temperature drift. Node 3 (external geometry self-scan): Node 5 provides a stability reference (low-motion + wear integrity) that improves scan quality scoring and repeatability labeling of derived geometry outputs. Node 4 (insertable compliance/distance/contact session device): Node 5 provides systemic state and motion gating that supports valid stimulus-response characterization and improved repeatability across sessions.

G. Privacy-Preserving Context Layer (Non-Limiting)

In various embodiments, Node 5 outputs are stored and displayed as abstract tags/indices (e.g., rest-state, high-motion, low-wear-integrity, elevated activation) and confidence indicators rather than raw ECG/PPG/EDA waveforms unless explicitly enabled. These context outputs may be used internally for system gating and normalization while preserving privacy.

H. Variations

Any embodiment described herein may vary form factor (chest, ring, wrist), sensor selection, thresholds, and algorithms while maintaining the functional intent of Node 5 providing upper-body context required to interpret intimate-region physiology as reliable, state-conditioned patterns.